AI-Augmented Project Controls for Large Capital Projects

A Decision Paper for Management and Supervisory Boards

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Contents

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Executive Summary

Large capital projects systematically underperform. Across a database of 16,000 projects, Oxford University researchers found that only 8.5 percent met their original cost and schedule targets — and a mere 0.5 percent delivered all promised benefits on time and on budget.[1] McKinsey's analysis of more than 300 projects exceeding one billion dollars in value found average cost overruns of 80 percent and schedule delays of 50 percent.[2] These are not outlier statistics; they describe the norm.

The root cause is not a shortage of engineering talent or project management methodology. It is an information problem: project controls data is generated too slowly, integrated too rarely, and interpreted too inconsistently for management to act before damage accumulates.

This paper proposes a practical, deployable response: a structured AI scaffolding system built on top of Claude Code — Anthropic's enterprise AI agent — that converts raw project data into structured cost performance analysis, escalation-ready reports, and visual dashboards in minutes rather than days. The system is not a replacement for experienced project professionals. It is a force multiplier: it ensures that earned value calculations are performed consistently every reporting period, that escalation thresholds are enforced automatically, and that do-

[1] Flyvbjerg, B. & Gardner, D. (2023). *How Big Things Get Done*. Macmillan. Database of 16,000 projects; 8.5% delivered on cost and schedule; 0.5% delivered all promised benefits.

[2] McKinsey & Company (2015). "Megaprojects: The good, the bad, and the better." Review of 300+ projects exceeding USD 1 billion.

main heuristics from decades of benchmarking are applied without relying on institutional memory.

The system has been designed and implemented. It is available as open-source code at github.com/jmeier1963/capital-project-management. A working prototype illustrating the approach on a hypothetical 500 km hydrogen transmission pipeline (EUR 1.75 billion CAPEX) demonstrates the full capability.

This paper asks the board to make three decisions:

1. **Authorize a pilot** on one live capital project, with a defined success criteria review at 90 days.
2. **Assign an owner** — a senior executive accountable for the pilot, sitting at the intersection of project controls, digital, and operations.
3. **Establish a data governance baseline** — define which project data is permissible to process through AI systems and under what conditions.

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1. The Persistent Performance Deficit in Large Capital Projects

1.1 The Scale of the Problem

Flyvbjerg's landmark study — the largest empirical analysis of project performance ever conducted — established what he calls the "iron law of megaprojects": over budget, over time, over and over again.[3] The headline numbers bear repeating for emphasis:

- **91.5%** of projects exceed their original budget or schedule, or both
- **8.5%** deliver within original parameters
- **0.5%** deliver cost, schedule, *and* promised benefits

McKinsey adds sectoral granularity. Rail projects overrun by an average of 45 percent. Bridges and tunnels by 35 percent. Mining and metals projects — among the most capital-intensive category — see 83 percent of major projects exceed planned CAPEX by more than 40 percent, with average delays of 20 to 30 months.[4]

The energy transition amplifies the stakes. The Hydrogen Council and McKinsey reported in 2025 that the global hydrogen sector has now committed over USD 110 billion across more than 500 projects past Final Investment Decision.[5] These are projects where schedule de-

[3] Flyvbjerg, B. (2017). "The Iron Law of Megaprojects." Cato Institute Policy Report. 91.5% of projects exceed budget, schedule, or both.

[4] McKinsey Global Institute (2016). "Reinventing Construction." Mining and metals sector overrun data. Rail, bridge, and tunnel statistics from Flyvbjerg's database.

[5] Hydrogen Council & McKinsey & Company (2025). *Global Hydrogen Compass*. USD 110B committed investment across 500+ post-FID projects.

lays translate directly into stranded capital, missed carbon commitments, and competitive disadvantage in a market where first-mover timing is commercially decisive.

1.2 Why Projects Fail: The Information Gap

The intuition that projects fail because of poor engineering or inadequate planning is only partially correct. McKinsey's analysis of 48 deeply troubled megaprojects found that **73 percent of cost and schedule overruns were caused by poor execution** — specifically, by failures of monitoring, escalation, and corrective action — not by flawed original designs.[6]

The mechanism is predictable. Project data — cost actuals, schedule progress, change orders, risk events — is generated continuously in the field. But in most organisations, this data travels slowly: from site time-sheets and contractor invoices into cost control systems, then into spreadsheets maintained by cost engineers, then into narrative reports assembled for management review. By the time a negative trend is visible in a board report, the underlying pattern may have been developing for three or four months.

The consequence is that corrective actions are taken late, when the cost of recovery is high. A schedule slip that costs EUR 2 million to correct in month three may cost EUR 20 million by month nine — not because the underlying problem grew tenfold, but because the window for low-cost intervention closed.

1.3 The Project Controls Baseline Problem

Consistent, rigorous project controls — earned value management (EVM), integrated schedule analysis, systematic risk quantification — are the established antidote. The Independent Project Analysis Group (IPA), whose proprietary database covers more than 20,000 capital projects, measures project performance against a Project Control Index (PCI). Projects with strong project controls consistently outperform their peers on cost, schedule, and operability outcomes.[7]

The barrier to universal adoption of rigorous project controls is not lack of awareness. It is capacity and consistency. A cost engineer maintaining full earned value analysis across eight WBS packages, three active contracts, and five reporting periods simultaneously — while also managing contractor queries, validating invoices, and preparing the monthly report — will, under pressure, abbreviate the analysis. Thresholds will not be checked. EAC forecasts will not be run across all three methods. The escalation that should have gone to the Project Director this week will go next week instead.

This is the specific gap that AI-augmented project controls addresses.

[6] McKinsey & Company (2017). "Increasing transparency in megaproject execution." Analysis of 48 troubled projects; 73% of overruns attributed to execution failures.

[7] Independent Project Analysis (IPA). *Capital Project System Improvement*. Project Control Index methodology and performance correlation. ipaglobal.com

2. What Has Changed: Why AI Can Help Now

2.1 The Structural Shift in AI Capability

For most of the past decade, AI in project management meant dashboards that required extensive data pipeline engineering, natural language processing tools that extracted text from documents with limited accuracy, and predictive models that required months of training on proprietary datasets before producing actionable output.

The shift that has occurred since 2023 is qualitative, not incremental. Large language models (LLMs) — and in particular AI agents that can reason over structured data, apply domain-specific rules, call tools, and generate structured outputs — have changed what is possible without bespoke development. An AI agent can today be given a CSV file of cost actuals, a set of rules encoded in plain language (e.g., "if CPI drops below 0.85, flag for Project Director escalation"), and a set of calculations to perform, and will produce a complete, consistent, correctly formatted report — every time, in under two minutes.

This is not a claim that AI understands project management in the way an experienced cost engineer does. It is a more precise and more practically important claim: AI can perform the *routine, structured, rule-governed portion* of project controls work with perfect consistency, freeing skilled professionals to focus on the *judgment-intensive* portions — contractor negotiation, root cause analysis, recovery planning — where human expertise is irreplaceable.

2.2 The Adoption Paradox — and How to Avoid It

Deloitte's 2025–2026 State of AI in the Enterprise survey found that while nearly 90 percent of companies have deployed AI in at least one business function, 94 percent report not seeing "significant" value from their investments.[8] This paradox has a structural explanation: most AI deployments in enterprises target knowledge work at the individual level (drafting emails, summarising documents) rather than the *process* level where value is concentrated.

In capital project controls, value is concentrated in process consistency, not individual productivity. The question is not whether a cost engineer can produce a report faster with AI assistance. The question is whether every project — regardless of the experience level of its controls team, regardless of whether it is month four or month thirty-two — produces a complete, consistently structured, threshold-checked EVM report on time, every reporting period.

The approach described in this paper is designed specifically for that target. It does not add an

[8] Deloitte US (2026). *State of AI in the Enterprise*. 89% of companies have deployed AI; 94% report not seeing significant value.

AI layer on top of existing processes. It embeds AI into the process as a non-optional execution step, with encoded domain rules, parameterised thresholds, and mandatory output formats that cannot be abbreviated under pressure.

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3. The System: Architecture and Capabilities

3.1 Design Principles

The system is built on four principles that distinguish it from generic AI implementations:

Data first, conversation second. Project intelligence is derived from structured data — CSV files, YAML configuration, JSON schemas — not from narratives or slide decks. This makes outputs reproducible, auditable, and comparable across periods.

Provenance on every heuristic. Cost benchmarks, productivity norms, and contingency ranges are stored as versioned YAML files with mandatory `source:` and `valid_range:` fields. When the AI applies a benchmark, it cites the source. When the project falls outside the valid range, it flags this explicitly.

Rules encoded, not remembered. Escalation thresholds, required approvals, and judgment rules (e.g., "any change order exceeding 1% of contract value triggers an independent cost review") are encoded in a `CLAUDE.md` configuration file — the project's AI constitution. They are applied automatically on every analysis run, not recalled from a briefing document.

Roles, not generics. Every communication or action recommendation produced by the AI addresses a specific named role from the project's organisation chart. It does not recommend that "the team" should act. It specifies that the Commercial Manager should review the change order, that the Project Director should receive the escalation memo, and that the cost register should be updated by the Cost Engineer within five business days.

3.2 System Components

The system consists of four integrated layers:

Layer 1 — Project Data Store

A structured directory of CSV, YAML, JSON, and Markdown files representing the project's living state: the project brief (scope, FID CAPEX, milestones), the contracts register, the WBS cost register, the resource matrix, and the risk register. All files have defined schemas that validate on intake. The directory is version-controlled, meaning every change is logged and reversible.

Layer 2 — Domain Heuristics Library

A curated library of parametric benchmarks and productivity norms, stored as versioned YAML files. For a hydrogen pipeline project these include: installed cost per kilometre by pipe diameter and terrain type, welding productivity norms per shift, compression station cost per megawatt, and contingency ranges by estimate class (AACE Class 1–5). Each file carries its source, valid range, and date of last update. The AI consults this library before generating any cost forecast and flags deviations that exceed ±30 percent of the benchmark.

Layer 3 — Specialist Agents

Three domain-specific agents that activate automatically on defined triggers:

Agent	Trigger	Primary Output
EVM Analyst	Monthly reporting cycle; any mention of cost or schedule performance	Full earned value report: five charts, RAG status, three EAC forecasts, escalation flags
Schedule Analyzer	Milestone slip; SPI below threshold; request for schedule review	Schedule health report with critical path float, recovery options matrix
Contract Reviewer	Change order received; claim notice; new contract for review	Structured commercial summary, flagged risk clauses, CO entitlement assessment
Risk Assessor	Monthly cycle; new risk identified; risk materialisation	EMV-ranked Top 10 risk digest, escalation flags, recommended register updates

Layer 4 — Governance Rules

Encoded in `CLAUDE.md`, the project's AI constitution. This file specifies mandatory escalation thresholds (CPI < 0.85, SPI < 0.85 for two consecutive periods), required approval levels for change orders, stage-gate-specific estimate class requirements, and output format standards. The AI cannot deviate from these rules; they are applied on every run without exception.

3.3 Integration with Existing Tools

The system reads Primavera P6 `.xer` schedule exports (via the P6XER MCP server), Excel and CSV cost exports, PDF contracts (via the PDF extraction skill), and Google Drive document repositories. It does not require replacing any existing tool. It adds a controlled intelligence layer on top of data that organisations are already generating.

4. The EVM Module: From Calculation to Institutional Discipline

4.1 What Earned Value Management Is — and Why It Is Underused

Earned Value Management is the internationally recognised standard (ANSI/EIA-748) for integrating cost and schedule performance measurement. Its core logic is simple: at any point in a project, you can compute not just what you have spent (Actual Cost, ACWP), but what you *should* have spent for the work you have actually completed (Earned Value, BCWP). The ratio of earned value to actual cost is the Cost Performance Index (CPI). A CPI of 0.91 means you are spending EUR 1.10 to deliver EUR 1.00 of budgeted work.

The power of EVM lies in its predictive validity. Research across thousands of projects has established that the CPI at the 20 percent completion milestone is a reliable predictor of final outcome. Projects that are underperforming at 20 percent completion rarely recover to budget — and when they do, it is because management intervened early, not because efficiency spontaneously improved.[9]

Despite this, EVM is performed inconsistently in practice. The calculation requires integrating cost actuals, progress measurements, and the original budget baseline — data that typically live in three or four separate systems and require manual reconciliation. Under schedule pressure, this reconciliation gets abbreviated. The result is that the single most powerful early warning indicator available to project management is produced late, inconsistently, or not at all.

4.2 What the EVM Module Delivers

Given a CSV file containing WBS codes, budget at completion (BAC), planned value (BCWS), earned value (BCWP), and actual cost (ACWP) — data that any functioning cost control system produces — the EVM module generates, in under two minutes:

- **A complete markdown report** with executive summary, WBS-level performance table with RAG status, three EAC forecasts (CPI method, composite CPI×SPI method, and planned-rate method), variance at completion, and TCPI
- **Five publication-ready charts**: S-curve (BCWS/BCWP/ACWP over time), CPI/SPI trend with threshold bands, WBS cost variance waterfall, EAC forecast comparison, and a CPI/SPI quadrant map with bubble sizes proportional to budget at stake
- **Automatic escalation flags** when CPI or SPI breach thresholds, when SPI has been below 0.85 for two consecutive periods, or when TCPI exceeds 1.10 (the level at which recovery is generally considered unrealistic without re-baselining)

[9] Christensen, D.S. (1993). "The Estimate at Completion Problem: A Review of Three Studies." *Project Management Journal.* CPI stability at 20% completion and its predictive validity for final outcome.

- **A plain-language interpretation** identifying the worst-performing package, the recommended EAC to use for board reporting, and the next three actions with named responsible parties

The recommended EAC for large capital projects is the composite CPI×SPI method, which assumes that schedule pressure will continue to drive cost efficiency downward. For the hydrogen pipeline example included with the system, this method produces an EAC of EUR 1.637 billion against an approved BAC of EUR 1.350 billion for the contracted scope — a 21 percent overrun signal at month five of a 36-month execution programme, early enough for effective intervention.

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5. Business Case

5.1 The Value Drivers

The business case for AI-augmented project controls operates on three distinct value levers:

Lever 1 — Earlier detection of adverse trends

Research by IPA establishes that the cost of corrective action increases approximately exponentially with the delay between signal and response. A schedule slip that can be corrected for EUR 2 million in month three may require EUR 15–20 million to recover by month nine, because interim procurement commitments, subcontractor mobilisation costs, and lost float compound the original deviation.

The system makes the trend visible in the first reporting period it appears, rather than the third. On a EUR 1 billion project, even a single such early intervention — preventing a 2 percent cost growth that would otherwise materialise — generates a return that exceeds the entire cost of implementing and operating the system.

Lever 2 — Consistent application of escalation rules

Escalation failures are documented as a primary driver of late-stage project crises. The mechanism is well understood: a cost engineer identifies a negative trend, judges it borderline, decides to wait one more period to confirm before escalating, and the window for low-cost intervention closes. Encoded escalation thresholds eliminate the discretion at the margin. When CPI falls below 0.85, the flag fires automatically. The Project Director is notified regardless of where the cost engineer's judgment falls.

Lever 3 — Reduced cost of producing controls outputs

Companies using AI-assisted project management tools report an average 15 percent improvement in project delivery productivity.[10] On the specific task of monthly reporting — which typically consumes 3–5 working days per reporting period for a major project's controls team — consistent findings indicate that AI assistance reduces this by 40–60 percent. This frees experienced project controls professionals to perform root cause analysis, contractor engagement, and recovery planning rather than data assembly and chart production.

5.2 Conservative Financial Model

The following model is intentionally conservative and uses a single EUR 1 billion project as the unit of analysis:

Value Driver	Basis	Annual Value (EUR)
Earlier detection — 1 avoided 2% cost growth	One intervention per project year	20,000,000
Reporting efficiency — 2 FTE-months freed per year	Senior engineer at EUR 15k/month fully loaded	360,000
Consistent escalation — avoided one late crisis	50% probability × EUR 5M average cost of late recovery	2,500,000
Total (conservative)		**~22,860,000**

Against this, the implementation costs are modest:

Cost Item	Basis	Annual Cost (EUR)
Claude Code enterprise licences (10 users)	Current enterprise pricing	~60,000
Internal implementation effort (one-time)	15 person-days × senior engineer rate	~30,000

[10] Various sources aggregated in: Celoxis (2025). "AI and Machine Learning in Project Management." 15% average productivity improvement; 61% on-time delivery with AI tools vs. 47% without.

Cost Item	Basis	Annual Cost (EUR)
Ongoing maintenance and data governance	0.25 FTE	~75,000
Total		**~165,000 p.a.**

The implied return on investment exceeds 100:1 on conservative assumptions, driven almost entirely by the value of a single avoided late-stage intervention. The ratio improves further on a portfolio of projects, where the fixed infrastructure cost (licences, governance, templates) is shared.

5.3 Contextual Benchmark

Companies that use AI-driven tools in project management deliver 61 percent of their projects on time, compared to 47 percent for those that do not — a 14-percentage-point improvement.[11] On a programme of five concurrent capital projects, each averaging EUR 500 million in CAPEX, moving one project from the "late" to "on-time" bucket — with a typical delay cost of 5–10 percent of CAPEX — represents EUR 25–50 million in value creation from the portfolio uplift alone.

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6. Implementation Roadmap

Phase 1 — Foundation (Months 1–3)

Objective: Install the system on one live project; establish data governance baseline; train the project controls team.

Activities: - Select pilot project: ideally a project at the 10–25 percent completion stage, with an active cost control team and at least three months of actuals history - Appoint an AI Project Controls Lead (existing staff member, not a new hire) - Establish data governance policy: define permissible data types, storage locations, and access controls for AI-processed project data - Install Claude Code; deploy the EVM skill and project scaffolding; load three months of historical cost data - Run the first EVM analysis; compare output to existing controls report; identify and close any data quality gaps - Conduct a one-day orientation for the project controls team and the Project Manager

Success gate: First AI-generated EVM report accepted by the Project Manager as the definitive monthly controls output.

[11] Ibid.

Phase 2 — Expansion (Months 4–9)

Objective: Extend to three projects; activate schedule and risk agents; integrate with existing reporting cadence.

Activities: - Replicate the deployment on two additional projects at different lifecycle stages (to stress-test the system across phases) - Activate the Schedule Analyzer and Risk Assessor agents - Connect to Primavera P6 exports (via P6XER MCP server) to automate schedule data ingestion - Integrate EVM report output into the monthly board reporting pack - Establish a quarterly heuristics review: update cost benchmarks and productivity norms based on project actuals

Success gate: At least one escalation flag correctly identified and acted upon ahead of the monthly Management Board review.

Phase 3 — Institutionalisation (Months 10–18)

Objective: Deploy across the full capital project portfolio; establish continuous improvement loop; build internal capability.

Activities: - Standardise the system as the mandatory project controls platform for all projects above a defined CAPEX threshold (recommended: EUR 50 million) - Develop project-type-specific heuristics libraries (pipeline, compression, offshore, civil infrastructure) - Establish a Centre of Excellence for AI-augmented project controls (2–3 FTE from existing staff, not a new department) - Publish a lessons-learned library: each project closeout contributes validated actuals back to the heuristics library, improving future estimates

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7. Risk Assessment and Governance

7.1 AI-Specific Risks

Risk: AI produces confident but incorrect analysis
Likelihood: Medium. Consequence: Medium.
The system mitigates this through schema validation, provenance requirements on all heuristics, and mandatory cross-checking of EAC against parametric benchmarks. No AI output is presented as final without a named human reviewer. The system generates recommendations; it does not make decisions.

Risk: Sensitive commercial data processed through third-party AI systems
Likelihood: Low with controls. Consequence: High.
Claude Code can be deployed on-premises or in a private cloud environment using

Anthropic's enterprise API. No project data need leave the organisation's controlled infrastructure. The data governance policy established in Phase 1 defines exactly which data categories are permissible.

Risk: Over-reliance reduces human expertise over time
Likelihood: Low with design. Consequence: Medium.
This risk is real and documented in analogous automation contexts. The mitigation is design: the system explicitly requires human interpretation of every report, ensures that escalation recommendations are reviewed not auto-executed, and generates plain-language explanations that build rather than bypass analytical understanding. The system is designed to augment the cost engineer's analytical output, not to replace the cost engineer.

7.2 Organisational Risks

Risk: Resistance from project controls professionals
Likelihood: High without change management. Consequence: Medium.
The most effective mitigation is framing. This system removes the least engaging portions of a cost engineer's work — data assembly, chart production, threshold checking — and returns time for the high-judgment work that experienced professionals value. Early engagement of senior project controls staff in the pilot design, and visible credit for the improved reporting outputs, is the primary change management lever.

Risk: Data quality too poor for meaningful analysis
Likelihood: Medium. Consequence: Medium.
The system will surface data quality problems that were previously obscured by manual report production. This is a feature, not a bug. The Phase 1 data quality review is specifically designed to establish a clean baseline before the system goes live on the pilot project.

Risk: Pilot succeeds but rollout stalls
Likelihood: Medium without board sponsorship. Consequence: High.
The pattern of successful pilot — stalled rollout — is the dominant failure mode for enterprise technology adoption. The mitigation is board-level ownership of the Phase 3 mandate and a defined CAPEX threshold above which the system is mandatory, not optional.

7.3 What This System Does Not Do

It is equally important to state explicitly what this system does not replace:

- It does not replace the Project Director's judgment on whether to escalate a situation to the board
- It does not negotiate with contractors, assess claim merits, or draft legal correspondence without human review
- It does not produce FID-quality cost estimates; it analyses cost *performance* against an

already-established baseline
- It does not replace the physical site inspection, quality review, or HSE incident investigation
- It does not guarantee project success; it improves the information available to the humans who determine outcome

Projects succeed because of skilled, experienced, well-led teams. This system makes it harder for those teams to miss early warning signals, easier to produce consistent reporting, and more likely that the right information reaches the right decision-maker at the right time.

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8. Recommendation and Board Decision Points

8.1 The Strategic Context

The hydrogen and energy transition investment wave is already underway. With USD 110 billion committed globally and project execution — not FID — now the bottleneck, the competitive advantage will accrue to organisations that execute consistently rather than those that plan ambitiously. Improving project controls quality from the industry median to the top quartile is worth, in expectation, 5–10 percent of total CAPEX across a portfolio. At scale, that is a strategic differentiator.

AI-augmented project controls is not a speculative technology bet. The EVM module described in this paper is working code, tested against real data, with verifiable outputs. The open-source release means the implementation risk is low: the system can be inspected in detail, extended without vendor dependency, and abandoned without sunk cost if the pilot does not demonstrate value.

The question for the board is not whether this technology works. It is whether the organisation is willing to commit the governance attention needed to deploy it at scale and embed it into the project controls process as a non-negotiable standard.

8.2 Three Decisions Required

Decision 1 — Authorize the pilot

Approve a 90-day pilot on one active capital project above EUR 100 million. Define three measurable success criteria in advance (suggested: first AI EVM report accepted as definitive by PM within 30 days; at least one escalation flag correctly generated; reporting time reduced by at least 30 percent).

Decision 2 — Assign executive ownership

Name a board member or direct report as executive sponsor. This is the single most important determinant of whether the pilot converts to rollout. The sponsor needs authority over the project controls function, visibility into the digital infrastructure, and a mandate to enforce the Phase 3 rollout if the pilot succeeds.

Decision 3 — Commission a data governance policy

Direct the Chief Information Officer or General Counsel (or equivalent) to produce a data governance policy covering AI-processed project data within 60 days. This policy should address data classification, permissible AI processing environments, and the human-review requirements for AI-generated outputs used in board reporting.

8.3 If the Pilot Succeeds

The Phase 3 mandate — system deployment on all capital projects above a defined CAPEX threshold — should be treated as a project controls standard, equivalent in status to the requirement to use Primavera P6 for scheduling or to conduct HAZOP for process safety. Optional adoption of a better standard invariably produces uneven quality. Mandatory adoption produces the institutional discipline that translates individual project improvements into portfolio performance.

The open-source code base is the starting point, not the end state. The organisation's own project actuals, accumulated over three to five years of deployment, will produce a proprietary heuristics library that reflects its specific asset types, geographies, and contractor base — a genuine competitive asset that cannot be replicated by organisations that did not start building it.

Appendix A — Key Metrics Produced by the EVM Module

Metric	Formula	What it tells management
CPI	BCWP ÷ ACWP	Cost efficiency: EUR earned per EUR spent
SPI	BCWP ÷ BCWS	Schedule efficiency: earned value vs plan
EAC (Composite)	ACWP + (BAC−BCWP) ÷ (CPI×SPI)	Final cost forecast, accounting for schedule pressure
VAC	BAC − EAC	Expected over- or under-run at completion
TCPI	(BAC−BCWP) ÷ (BAC−ACWP)	CPI required for all remaining work to finish within budget; >1.10 indicates re-baselining is needed

RAG thresholds (industry standard):

Index	GREEN	AMBER	RED
CPI	≥ 0.95	0.85 – 0.94	< 0.85
SPI	≥ 0.95	0.85 – 0.94	< 0.85

Appendix B — Technical Resource

The working system — EVM skill, scaffolding templates, domain heuristics, JSON schemas, specialist agent definitions, and a complete example project — is available at:

https://github.com/jmeier1963/capital-project-management

The repository includes: - `evm/evm_calculator.py` — the full Python engine (713 lines, no external dependencies beyond pandas and matplotlib) - `scaffolding/` — project configuration templates for immediate deployment - `examples/H2-PIPE-DE-001/` — a complete worked example for a 500 km hydrogen pipeline, including five months of EVM data and the resulting analysis outputs - `README.md` — step-by-step installation and usage instructions

Installation requires copying three files and running one `pip install` command.

References

This paper was prepared to support a board-level decision on the adoption of AI-augmented project controls. It describes a working system and is accompanied by verifiable open-source code. The financial projections are illustrative and conservative; project-specific analysis should be conducted before committing to a scaled rollout.